Supp threshold = 3

Ex 2

The given tree does NOT contain a single path.

Cond. FP-tree on {a} | 7 ⑤

① {a}:7

① {a, c, f}:3
{a, b, c}:4

⑥ ~~{a, c, f}~~:3
~~{a, c, b}~~:4

② ✗

② item | freq.
a | 7
b | 4
c | 7
f | 3

③ item | freq.
a | 7
b | 4
c | 7
f | 3

④ item | freq.
a | 7
c | 7
b | 4
f | 3

⑦ item | head of node-link
c → C:7
b → f:3, b:4
f →

✗ single path

Cond FP-tree on {f, a} | 3 ⑤

① {f,a}:3 path

① {c, f}:3

⑥ ~~{c, f}~~:3

② {f, a, c}:3

② item | freq.
c | 3
f | 3

③ item | freq
c | 3
f | 3

④ item | freq
c | 3
f | 3

⑦ item | head of ..
c → C:3

Cond FP-tree on {b, a} | 4

① {b, a}:4 single path

② {b, a, c}:4

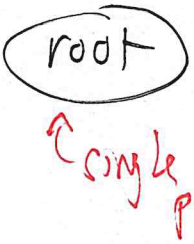
item | head of node-link
c →

C:4

single path

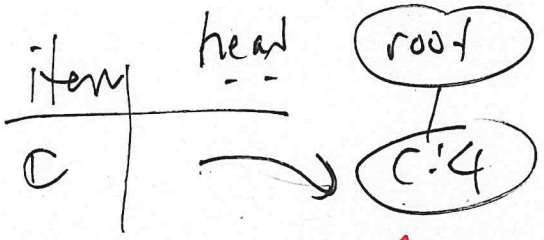
Cond. FP-tree on {c,a} 7

① [c,a]: 7
② X



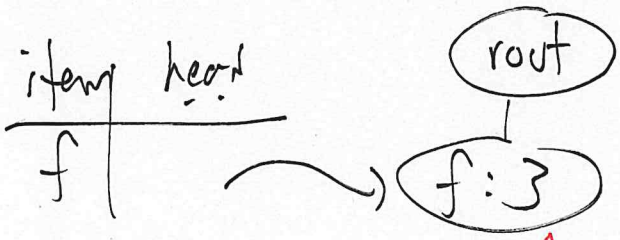
Cond. FP-tree on {b} 8

① {b}: 8
② [b,c]: 4



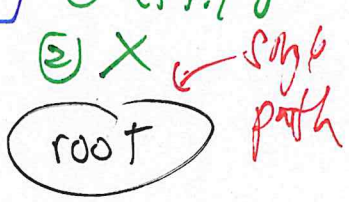
Cond. FP-tree on {c} 9

① [c]: 9
② [c,f]: 3



Cond. FP-tree on {f} 10

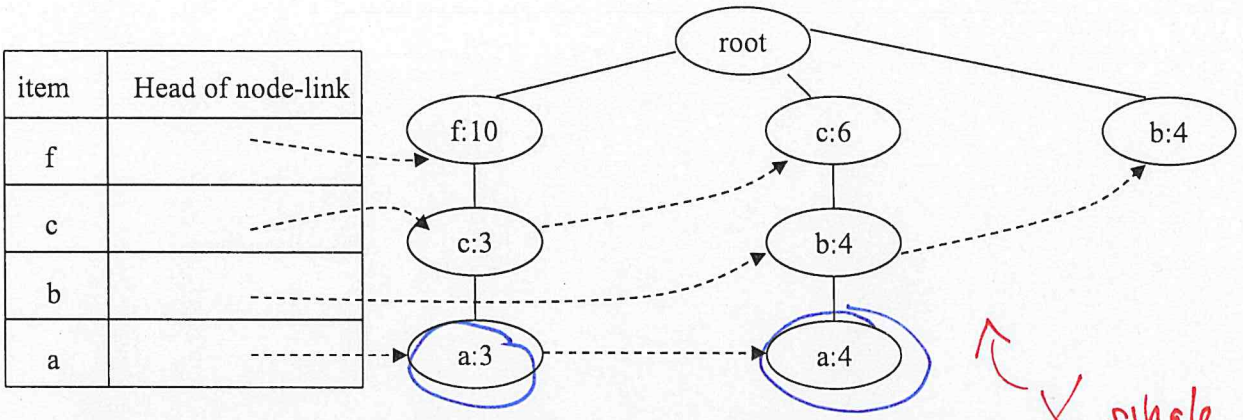
① {f}: 10
② X



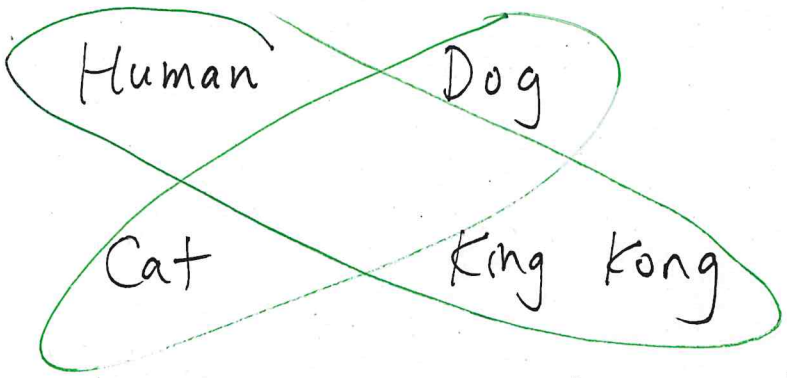
freq. patterns = { {a}: 7, {f,a}: 3, {f,a,c}: 3
{b,a}: 4, {b,a,c}: 4, {c,a}: 7
{b}: 8, {b,c}: 4, {c}: 9, {c,f}: 3
{f}: 10 }

COMP5331 Knowledge Discovery in Databases (Fall Semester 2026)
Exercise 2
FP-tree

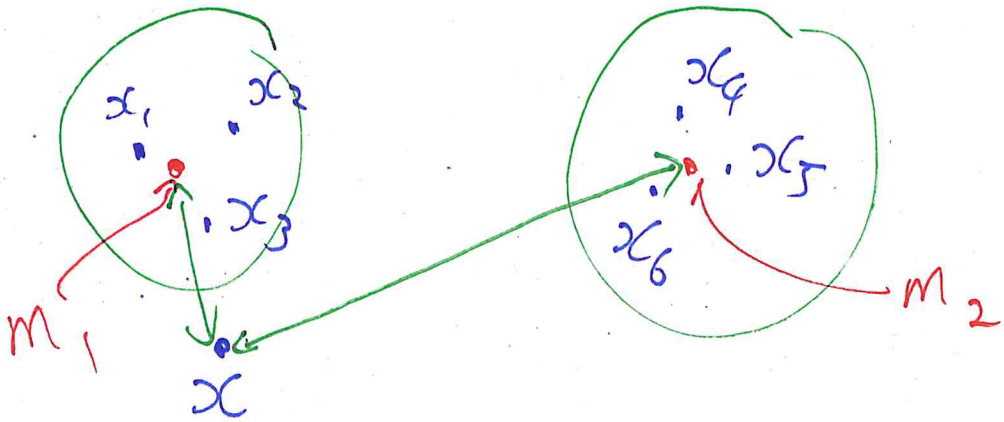
Q1 FP-tree



Consider the above FP-tree T and given the support threshold of 3.
Apply the algorithm of FP-growth(T, NULL) and generate all the conditional FP-trees.
What are the frequent patterns generated?



	Pet	No. of legs	No. of tails
Human	N	2	0
Dog	Y	4	1
Cat	Y	4	1
King Kong	N	2	0



e.g. $x_1 : (1, 2)$

$x_2 : (3, 4)$

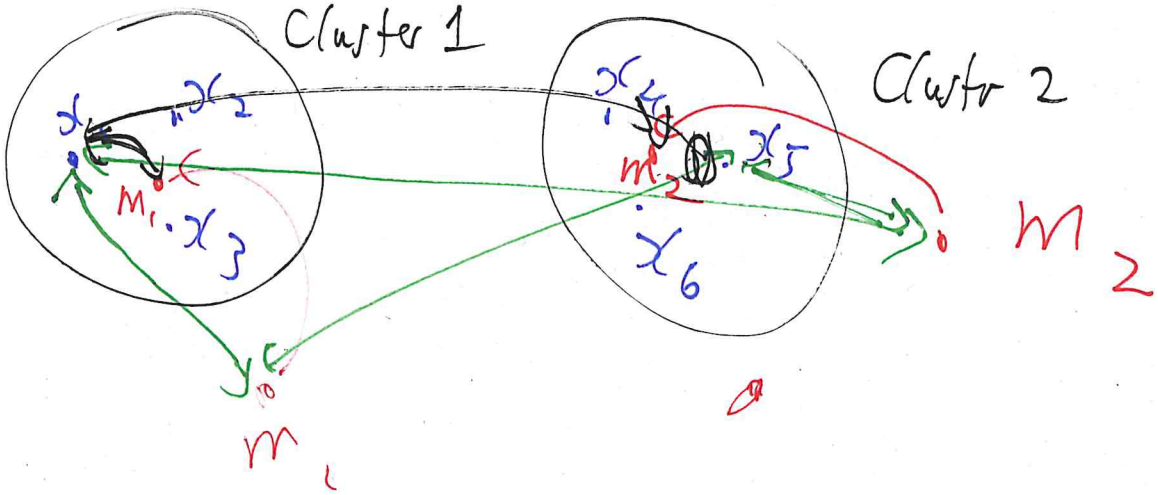
$x_3 : (5, 6)$

mean

$$m_1 = \frac{x_1 + x_2 + x_3}{3}$$

$$= \frac{(1,2) + (3,4) + (5,6)}{3}$$

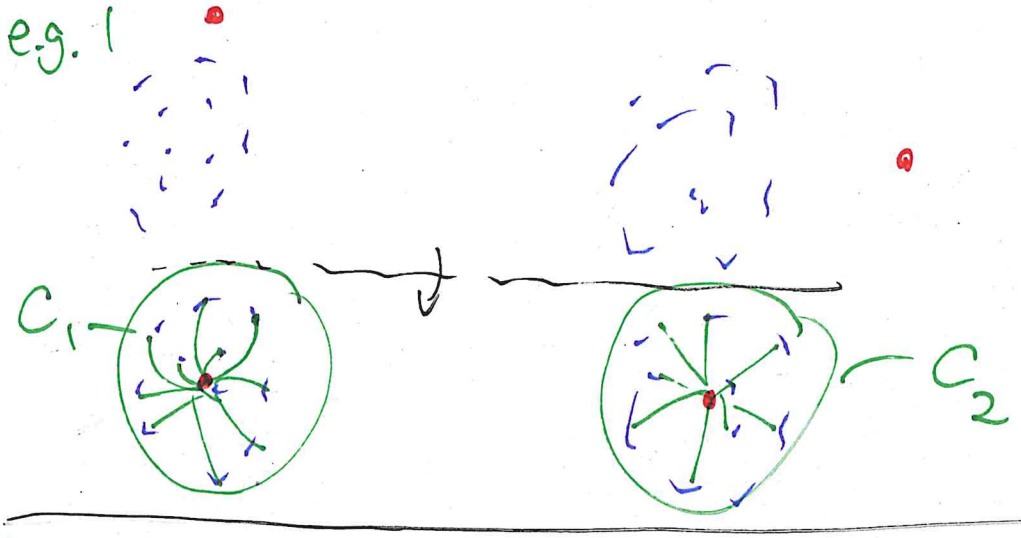
$$= \left(\frac{1+3+5}{3}, \frac{2+4+6}{3} \right)$$


Cluster 1
Cluster 2
x_1
x_2
m_1
x_3
x_4
m_2
x_5
x_6
m_2
m_1

goodness of clustering

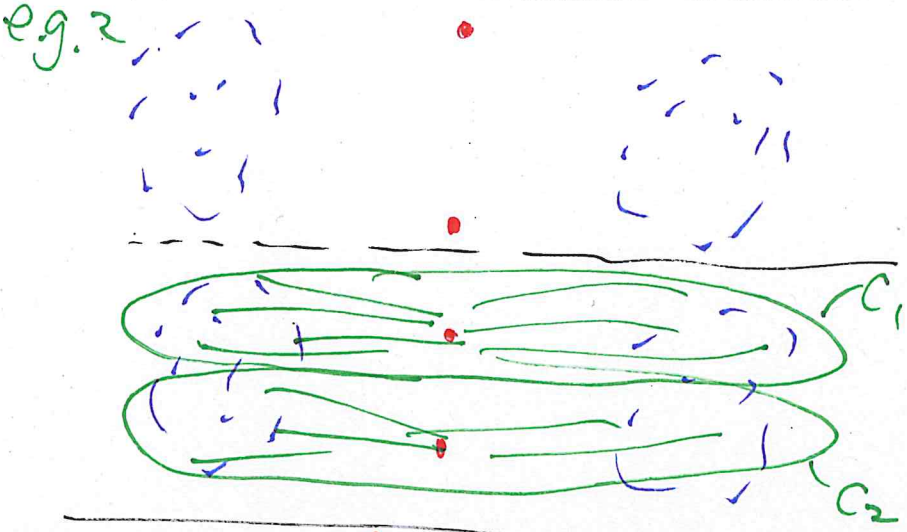
↑ avg. dist. bet. a point and its nearest mean

e.g. 1



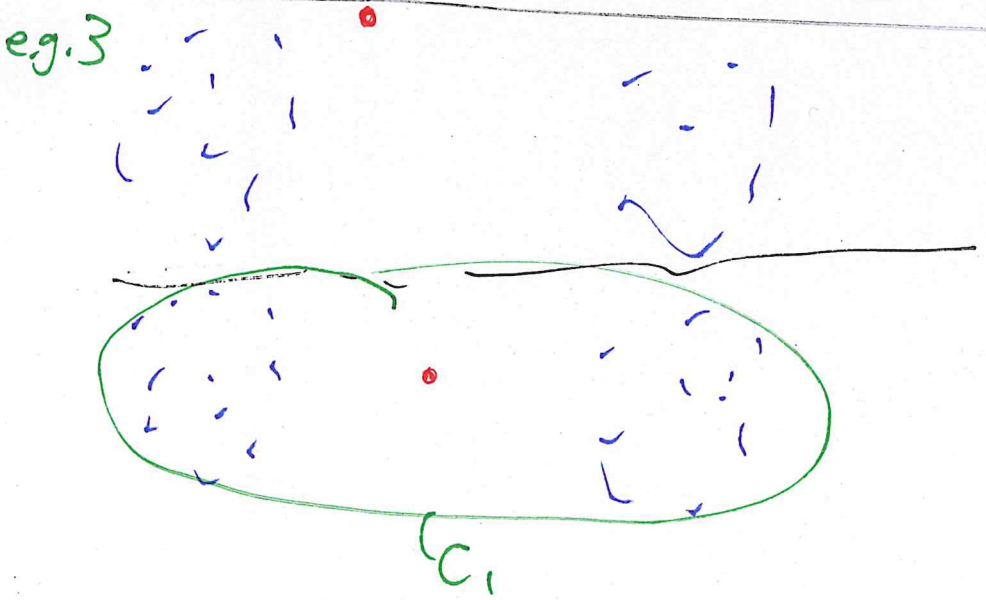
7.2

e.g. 2

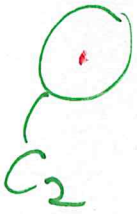


19.9

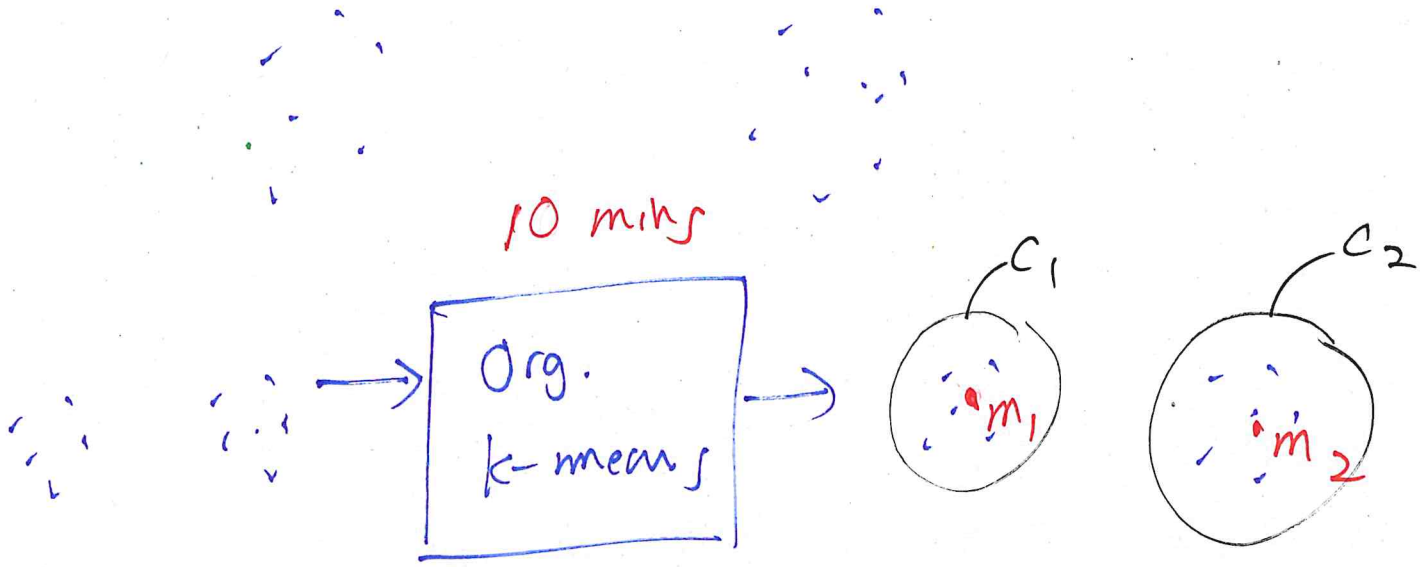
e.g. 3



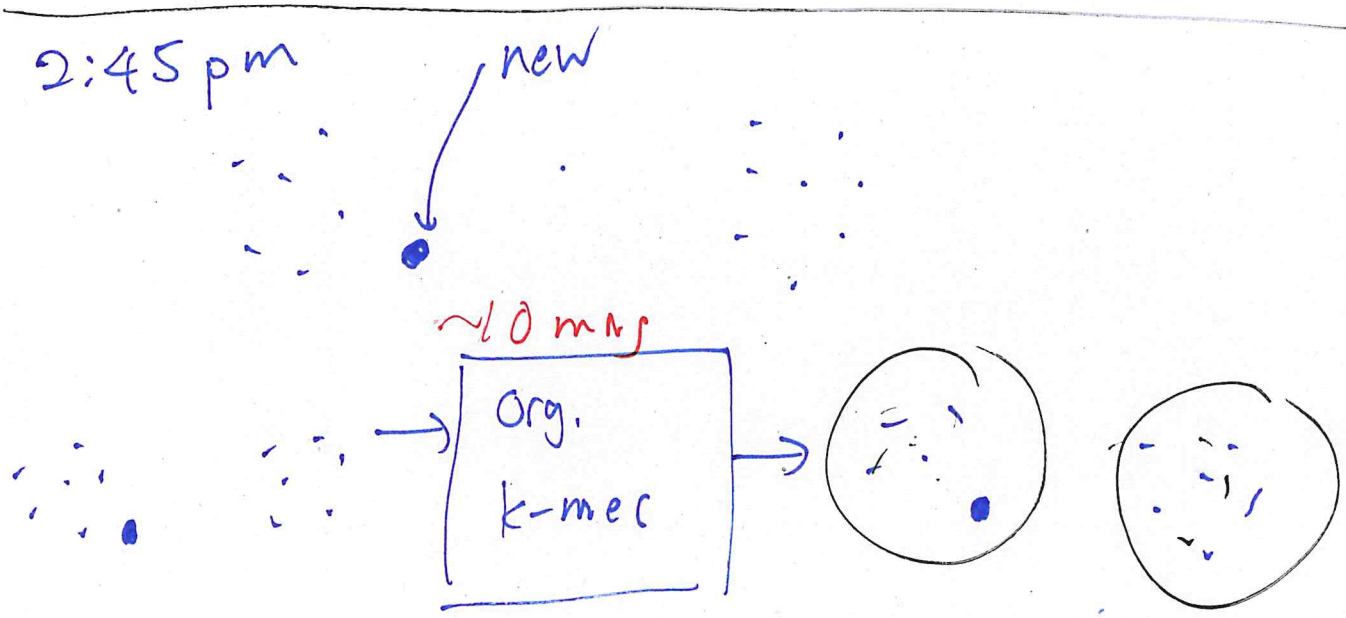
20.1

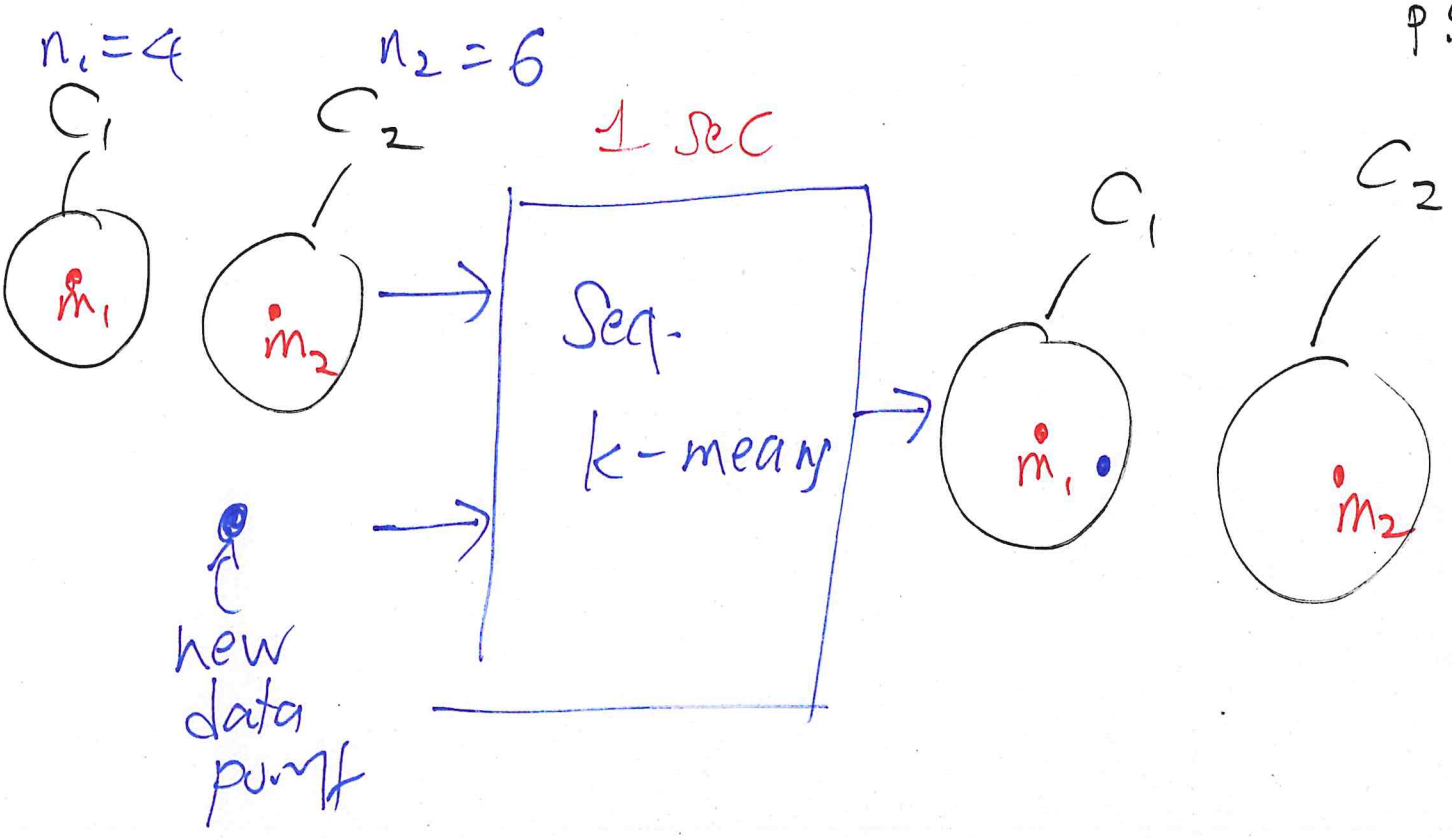


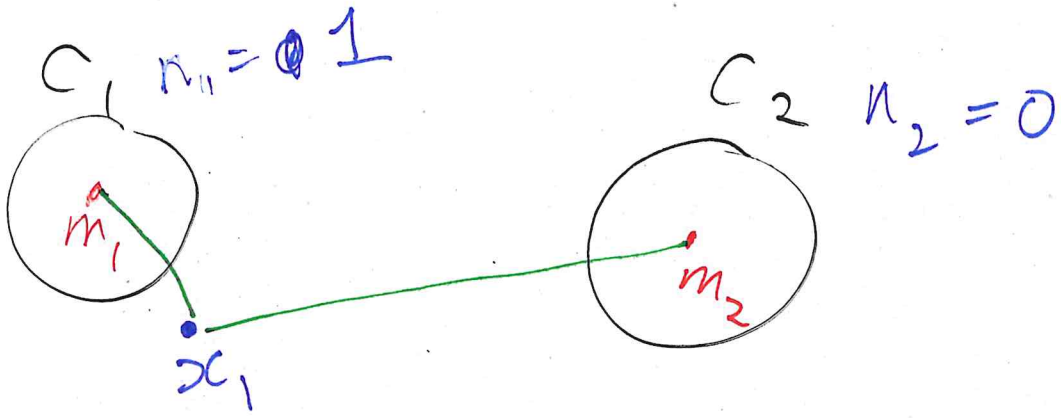
2:43 pm



2:45 pm



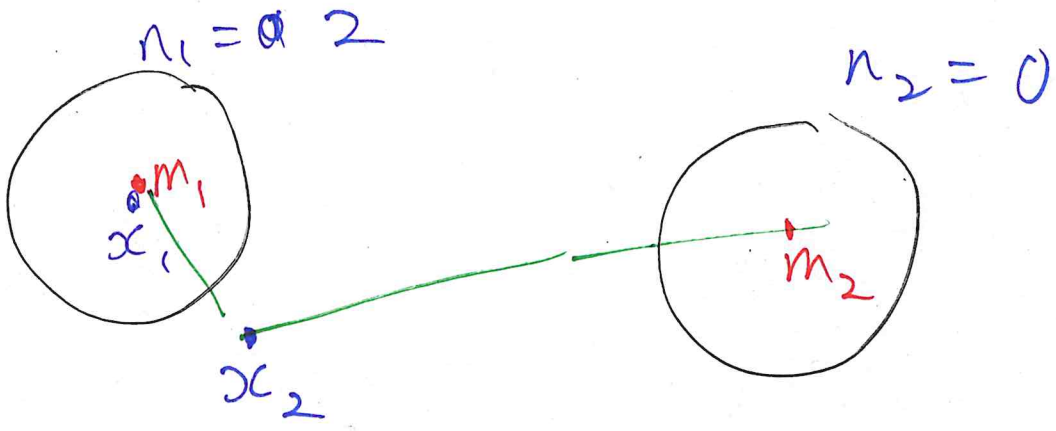

$n_1 = 4$
$n_2 = 6$
C_1
C_2
m_1
m_2
1 sec
Seq.
k-means
C_1
C_2
m_1
m_2
new
data
pt



$$m_1 \longleftarrow m_1 + \frac{1}{n_1}(x_1 - m_1)$$

$$= m_1 + \frac{1}{1}(x_1 - m_1)$$

$$= x_1$$



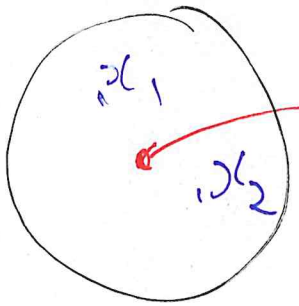
$$m_1 \leftarrow m_1 + \frac{1}{n_1}(x_2 - m_1)$$

$$= m_1 + \frac{1}{2}(x_2 - m_1)$$

$$= x_1 + \frac{1}{2}(x_2 - x_1)$$

$$= \frac{2x_1 + x_2 - x_1}{2}$$

$$= \frac{x_1 + x_2}{2}$$



$$\text{mean} = \frac{x_1 + x_2}{2}$$